Ryosuke Hirai

Curriculum Vitae

Personal Information

Name	Ryosuke Hirai (平井 遼介)
Nationality	Japanese
Gender	Male
Date of Birth	12th August, 1989
Email	ryosuke.hirai@riken.jp, ryosuke.hirai@monash.edu

Brief summary

Theoretical and computational astrophysicist specialising in binary evolution, compact objects, supernovae, and astrophysical transients. Combines analytical modelling, hydrodynamical simulations, stellar evolution calculations, and few-body dynamics to connect stellar evolution with multi-messenger observations.

Research Interests

2012–	Binary stars and supernovae
2015–	Efficient numerical schemes for difference equations
2018–	Stellar mergers and triple dynamics
2021–	Wind accretion in close binaries
2024–	Neutron star rocket model

Education

2014–2017	**Doctor of Science**, *Waseda University*, Advanced Research Institute of Science and Engineering.
2012–2014	**Master of Science**, *Waseda University*, Advanced Research Institute of Science and Engineering.
2008–2012	**Bachelor of Engineering**, *Waseda University*, School of Advanced Science and Engineering.

Research Experience

2024.3–	**Special Postdoctoral Research Fellow**, *at Astrophysical Big Bang Laboratory*, RIKEN.
2019.12–	**Research Fellow**, *at School of Physics and Astronomy*, Monash University.
2017–2019	**JSPS Overseas Research Fellow**, *at Department of Physics*, University of Oxford.

2017.4–10 **JSPS Research Fellow (PD)**, *at Advanced Research Institute of Science and Engineering*, Waseda University.

2016–2017 **JSPS Research Fellow (DC2)**, *at Advanced Research Institute of Science and Engineering*, Waseda University.

Teaching Experience

2026.5 **One-day Lecturer**, *PHS1011x*, Monash University.

2025.5 **One-day Lecturer**, *PHS1011x*, Monash University.

2024.8 **One-day Lecturer**, *PHS1022x*, Monash University.

2024.6 **Teaching Assistant**, *MESA Summer school*, University of Sydney.

2021–2022 **Facilitator**, *Astrophysics Book Club*, Monash University.

2020.7–10 **Teaching Assistant**, *PHS1022 (Waves and Quantum Physics)*, Monash University.

2014–2017 **Teaching Assistant**, *Introductory Physics*, Waseda University.

2012–2014 **Cram School Teacher**, *Junior High School Maths*, Shingakusha.

2008–2016 **Cram School Teacher**, *Primary to Junior High School Maths*, Ikuho Shingakujuku.

Research Supervision

PhD students

Co-supervised 5 PhD students.

Honours students

Supervised 1 Honours student and co-supervised 2.

Undergraduate students

Supervised 1 student for their undergraduate thesis, 2 students for a research course and 5 students for summer/winter vacation projects.

Grants and Awards

2026 **UJA Young Investigator Award**.

2025 **OzGrav Mentor Award**, *Highly Commended*.

2024–2027 **RIKEN Special Postdoctoral Research Fellow, Grant**, *1000000JPY $\times 3$ yr + 1000000 JPY (extra)*.

2023 **OzGrav Scientific Achievement Award–Astronomy Theme**, *Winner*.

2023 **OzGrav Rising Star Award–Postdoc**, *Runner-up*.

2023 **Best Presentation Prize (1st place), Monash Science ECN Symposium**, *2000 AUD*.

2018 **Hayakawa Satio Fund, Astronomical Society of Japan**, *~210000 JPY*.

2016–2017 **JSPS Research Fellow (DC2), Grant**, *1200000+1100000 JPY*.

2015 **Research Grant for Young Scientists, Early Bird Program from Waseda Research Institute for Science and Engineering**, *400000 JPY*.

Time Allocation

2025 **HST GO Cycle 33 (PI: Qinan Wang)**, *9 orbits.*

2025 **Monash Partner Share Allocation Scheme Q2, National Computational Infrastructure (NCI) Australia**, *880 KSU on NCI Gadi Supercomputer.*

2025 **Monash Partner Share Allocation Scheme Q1, National Computational Infrastructure (NCI) Australia**, *300 KSU on NCI Gadi Supercomputer.*

2025 **ADACS Merit Allocation Program**, *12 weeks development support for optimizing HORMONE (≈42000 AUD).*

2024 **MeerKAT Open Time (PI: Rob Fender)**, *Grade A.*

2024 **ADACS Merit Allocation Program**, *12 weeks development support for MPI-parallelizing HORMONE (≈42000 AUD).*

2023 **Adapter Allocation Scheme Q2, National Computational Infrastructure (NCI) Australia**, *208 KSU on NCI Gadi Supercomputer.*

2023 **ANU 2.3 m Telescope (PI: Ryosuke Hirai)**, *Priority band 2, 9 dark hours.*

2017 **ESO VLT MUSE (PI: Frédéric Vogt)**, *C class.*

Publications

First Author

[14] **Supernova-induced binary-interaction-powered supernovae: a model for SN2022jli**
Ryosuke Hirai, Philipp Podsiadlowski, Peter Hoeflich, Maxim V. Barkov, Conrad Chan, David Liptai, Shigehiro Nagataki
The Astrophysical Journal, Volume 995, Issue 1, id.55, 17 pp. (2025)

[13] **Neutron star kicks plus rockets as a mechanism for forming wide low-eccentricity neutron star binaries**
Ryosuke Hirai, Philipp Podsiadlowski, Alexander Heger, Hiroki Nagakura
The Astrophysical Journal Letters, Volume 972, Issue 1, id.L18, 8 pp. (2024)

[12] **Constraining mass-transfer and common-envelope physics with post-supernova companion monitoring**
Ryosuke Hirai
Monthly Notices of the Royal Astronomical Society, Volume 523, Issue 4, pp.6011-6019 (2023)

[11] **A two-stage formalism for common-envelope phases of massive stars**
Ryosuke Hirai, Ilya Mandel
The Astrophysical Journal Letters, Volume 937, Issue 2, id.L42, 7 pp. (2022)

[10] **Neutron stars colliding with binary companions: formation of hypervelocity stars, pulsar planets, bumpy superluminous supernovae and Thorne-Żytkow objects**
Ryosuke Hirai, Philipp Podsiadlowski
Monthly Notices of the Royal Astronomical Society, Volume 517, Issue 3, pp.4544-4556 (2022)

[9] **Conditions for accretion disc formation and observability of wind-accreting X-ray binaries**
Ryosuke Hirai, Ilya Mandel
Publications of the Astronomical Society of Australia, Volume 38, article id. e056 (2021)

[8] **Simulating the formation of η Carinae's surrounding nebula through unstable triple evolution and stellar merger-induced eruption**
Ryosuke Hirai, Philipp Podsiadlowski, Stanley Owocki, Fabian R. N. Schneider, Nathan Smith
Monthly Notices of the Royal Astronomical Society, Volume 503, Issue 3, pp.4276-4296 (2021)

[7] **Formation pathway for lonely stripped-envelope supernova progenitors: implications for Cassiopeia A**
Ryosuke Hirai, Toshiki Sato, Philipp Podsiadlowski, Alejandro Vigna-Gómez, Ilya Mandel
Monthly Notices of the Royal Astronomical Society, Volume 499, Issue 1, pp.1154-1171 (2020)

[6] **Comprehensive study of ejecta-companion interaction for core-collapse supernovae in massive binaries**
Ryosuke Hirai, Philipp Podsiadlowski, Shoichi Yamada
The Astrophysical Journal, Volume 864, Issue 2, article id. 119, 17 pp. (2018)

[5] **The Origin of the Possible Massive Black Hole in the Progenitor System of iPTF13bvn**
Ryosuke Hirai
Monthly Notices of the Royal Astronomical Society: Letters, Volume 469, Issue 1, p.L94-L98 (2017)

[4] **Formation Scenario of the Progenitor of iPTF13bvn Revisited**
Ryosuke Hirai
Monthly Notices of the Royal Astronomical Society, Volume 466, Issue 4, p.3775-3783 (2017)

[3] **Hyperbolic Self-Gravity Solver for Large Scale Hydrodynamical Simulations**
Ryosuke Hirai, Hiroki Nagakura, Hirotada Okawa, Kotaro Fujisawa
Physical Review D, Volume 93, Issue 8, article id.083006 (2016)

[2] **Possible Signatures of Ejecta-Companion Interaction in iPTF 13bvn**
Ryosuke Hirai, Shoichi Yamada
The Astrophysical Journal, Volume 805, Issue 2, article id. 170, 7 pp. (2015)

[1] **The Outcome of Supernovae in Massive Binaries; Removed Mass, and its Separation Dependence**
Ryosuke Hirai, Hidetomo Sawai, Shoichi Yamada
The Astrophysical Journal, Volume 792, Issue 1, article id. 66, 15 pp. (2014)

Co-Author

[45] **The Best Guess: Testing new and old formalisms for the common envelope against observations**
Riley Thai, Amanda Karakas, Zara Osborn, Robert Izzard, Ryosuke Hirai, Alex Kemp, Simon Campbell
submitted to ApJ

[44] **Dust Formation in Common Envelope Binary Interactions — III. Lightcurves**
Chunliang Mu, Orsola De Marco, Luis C. Bermúdez-Bustamante, Ryosuke Hirai, Daniel J. Price, Lionel Siess, Miguel González-Bolívar, Mike Y. M. Lau, and Nadejda Blagorodnova
submitted to MNRAS

[43] **From mass-loss histories to lightcurves: a generalised framework for interaction-powered transients**
Nikhil Sarin, Ryosuke Hirai
submitted to MNRAS

[42] **Comprehensive neutrino light curves and spectra: from pre-supernova evolution to early supernova phase**
Chinami Kato, Hiroki Nagakura, Akira Ito, Ryosuke Hirai, Shun Furusawa, Takashi Yoshida, Ryuichiro Akaho
submitted to ApJS

[41] **Mass transfer stability for AM CVn binaries with white dwarf donors**
Lucy O. McNeill, Ryosuke Hirai
Proceedings of Science, 87th Fujihara Seminar: The 50th Anniversary Workshop of the Disk Instability Model in Compact Binary Stars (DIM50th2025), 46

[40] **Weakened Inspirals I: High Mass Ratio Common Envelope Interactions in RGB Stars**
Jack Nibbs, Orsola De Marco, Lionel Siess, Ryosuke Hirai, Daniel Price
Publications of the Astronomical Society of Australia, Volume 43, id.e066, 21 pp. (2026))

[39] **Evolution of Massive Main-sequence Stars in Rapid Population Synthesis. I. Framework and Implementation**
Adam Brček, Ryosuke Hirai, Ilya Mandel, Harmony Lower
The Astrophysical Journal, Volume 1002, Issue 1, id.78, 13 pp. (2026)

[38] **Impacts of hydrogen envelope on supernova fallback and the resulting compact remnant masses**
Kengo Shinoda, Yudai Suwa, Ryosuke Hirai, Ryo Sawada, Kengo Tomida, Kazunari Iwasaki, Takeru K. Suzuki
accepted for publication in MNRAS

[37] **Delayed Wind Onset in Pa 30, the Remnant of Type Iax SN 1181**
Takatoshi Ko, Ryosuke Hirai, Taiga Sasaoka, Toshikazu Shigeyama
Publications of the Astronomical Society of Japan, Volume 78, Issue 2, pp. 645-654, 10 pp. (2026)

[36] **Tidal heating in detached double white dwarf binaries**
Lucy O. McNeill, Ryosuke Hirai
The Astrophysical Journal, Volume 992, Issue 1, id.154, 18 pp. (2025)

[35] **Rapid stellar and binary population synthesis with COMPAS: methods paper II**
Ilya Mandel, Jeff Riley, Adam Boesky, Adam Brcek, Ryosuke Hirai, Veome Kapil, Mike Y. M. Lau, JD Merritt, Nicolás Rodríguez-Segovia, Isobel Romero-Shaw, Yuzhe Song, Simon Stevenson, Avi Vajpeyi, L. A. C. van Son, Alejandro Vigna-Gómez, Reinhold Willcox
The Astrophysical Journal Supplement Series, Volume 280, Issue 1, id.43, 9 pp. (2025)

[34] **Population Synthesis Study on the Binary Origin of Type Ibn Supernovae**
Takatoshi Ko, Tomoya Kinugawa, Daichi Tsuna, Ryosuke Hirai, Yuki Takei
Monthly Notices of the Royal Astronomical Society, Vol. 541, Issue 4, pp. 3747-3762 (2025)

[33] **Common envelopes in massive stars III: The obstructive role of radiation transport in envelope ejection**
Mike Y. M. Lau, Ryosuke Hirai, Daniel J. Price, Ilya Mandel, Matthew R. Bate
Astronomy & Astrophysics, Volume 699, id.A274, 18 pp. (2025)

[32] **Precursor Activity Preceding Interacting Supernovae I: Bridging the Gap with SN 2022mop**
S. J. Brennan et al. (including R. Hirai)
submitted to A&A

[31] **Evidence of non-Solar elemental composition in the clocked bursts from SRGA J144459.2–604207**
Akira Dohi, Nobuya Nishimura, Ryosuke Hirai, Tomoshi Takeda, Wataru Iwatari, Toru Tamagawa, Amira Aoyama, Teruaki Enoto, Satoko Iwata, Yo Kato, Takao Kitaguchi, Tatehiro Mihara, Naoyuki Ota, Takuya Takahashi, Sota Watanabe, Kaede Yamasaki
Publications of the Astronomical Society of Japan, Volume 77, Issue 3, pp.L17-L23 (2025)

[30] **NinjaSat monitoring of Type-I X-ray bursts from the clocked burster SRGA J144459.2–604207**
Tomoshi Takeda et al. (including Ryosuke Hirai)
Publications of the Astronomical Society of Japan, Volume 77, Issue 3, pp.L24-L30 (2025)

[29] **Constraining the CSM structure and progenitor mass-loss history of interacting supernovae through 3D hydrodynamic modeling: The case of SN 2014C**
Salvatore Orlando, Emanuele Greco, Ryosuke Hirai, Tomoki Matsuoka, Marco Miceli, Shigehiro Nagataki, Masaomi Ono, Ke-Jung Chen, Dan Milisavljevic, Daniel Patnaude, Fabrizio Bocchino, Nancy Elias-Rosa
The Astrophysical Journal, Volume 977, Issue 1, id.118, 35 pp. (2024)

[28] **Evolution of the Convective Core Mass during the Main Sequence**
Minori Shikauchi, Ryosuke Hirai, Ilya Mandel
The Astrophysical Journal, Volume 984, Issue 2, id.149, 10 pp. (2025)

[27] **Bumpy Superluminous Supernovae Powered by Magnetar-star Binary Engine**
Jin-Ping Zhu, Liang-Duan Liu, Yun-Wei Yu, Ilya Mandel, Ryosuke Hirai, Bing Zhang, Aming Chen
The Astrophysical Journal Letters, Volume 970, Issue 2, id.L42, 10 pp. (2024)

[26] **Fits for the convective envelope mass in massive stars**
Lewis Picker, Ryosuke Hirai, Ilya Mandel
The Astrophysical Journal, Volume 969, Issue 1, id.1, 8 pp. (2024)

[25] **Expansion of accreting main-sequence stars during rapid mass transfer**
Mike Y. M. Lau, Ryosuke Hirai, Ilya Mandel, Christopher A. Tout
The Astrophysical Journal Letters, Volume 966, Issue 1, id.L7, 10 pp. (2024)

[24] **Dust formation in common envelope binary interactions – II: 3D simulations with self-consistent dust formation**
Luis C. Bermúdez-Bustamante, Orsola De Marco, Lionel Siess, Daniel J. Price, Miguel González-Bolívar, Mike Y. M. Lau, Chunliang Mu, Ryosuke Hirai, Taïssa Danilovich, Mansi M. Kasliwal
Monthly Notices of the Royal Astronomical Society, Volume 533, Issue 1, pp.464-481 (2024)

[23] **Evidence for Evolved Stellar Binary Mergers in Observed B-type Blue Supergiants**
Athira Menon, Andrea Ercolino, Miguel A. Urbaneja, Daniel J. Lennon, Artemio Herrero, Ryosuke Hirai, Norbert Langer, Abel Schootemeijer, Emmanouil Chatzopoulos, Juhan Frank, Sagiv Shiber
The Astrophysical Journal Letters, Volume 963, Issue 2, id.L42, 12 pp. (2024)

[22] **SN 2022jli: a type Ic supernova with periodic modulation of its light curve and an unusually long rise**
Moore et al. (including Ryosuke Hirai)
The Astrophysical Journal Letters, Volume 956, Issue 1, id.L31, 13 pp. (2023)

[21] **The Impact of Angular Momentum Loss on the Outcomes of Binary Mass Transfer**
Reinhold Willcox, Morgan MacLeod, Ilya Mandel, Ryosuke Hirai
The Astrophysical Journal, Volume 958, Issue 2, id.138, 12 pp. (2023)

[20] **Time-independent Simulations of Steady-State Accretion with Nuclear Burning**
Kaho Tse, Alexander Heger, Ryosuke Hirai, Duncan Galloway
submitted to ApJ

[19] **Rapid population synthesis of black-hole high-mass X-ray binaries: implications for binary stellar evolution**
Isobel M. Romero-Shaw, Ryosuke Hirai, Arash Bahramian, Reinhold Willcox, Ilya Mandel
Monthly Notices of the Royal Astronomical Society, Volume 524, Issue 1, pp.245-259 (2023)

[18] **The Galactic underworld: The spatial distribution of compact remnants**
David Sweeney, Peter Tuthill, Sanjib Sharma, Ryosuke Hirai
Monthly Notices of the Royal Astronomical Society, Volume 516, Issue 4, pp.4971-4979 (2022)

[17] **Common envelopes in massive stars II: The distinct roles of hydrogen and helium recombination**
Mike Y. M. Lau, Ryosuke Hirai, Daniel J. Price, Ilya Mandel
Monthly Notices of the Royal Astronomical Society, Volume 516, Issue 4, pp.4669-4678 (2022)

[16] **Common envelope binary interaction simulations between a thermally-pulsating AGB star and a low mass companion**
Miguel González-Bolívar, Orsola De Marco, Mike Y. M. Lau, Ryosuke Hirai, Daniel J. Price
Monthly Notices of the Royal Astronomical Society, Volume 517, Issue 3, pp.3181-3199 (2022)

[15] **An environmental analysis of the Type Ib SN 2019yvr and the possible presence of an inflated binary companion**
Ning-Chen Sun, Justyn R. Maund, Paul Crowther, Ryosuke Hirai, Amir Kashapov, Ji-Feng Liu, Liang-Duan Liu, Emmanouil Zapartas
Monthly Notices of the Royal Astronomical Society, Volume 510, Issue 3, pp.3701-3715 (2022)

[14] **Common envelopes in massive stars: The role of radiation pressure and recombination energy in ejecting red supergiant envelopes**
Mike Lau, Ryosuke Hirai, Miguel González-Bolívar, Daniel J. Price, Orsola De Marco and Ilya Mandel
Monthly Notices of the Royal Astronomical Society, Volume 512, Issue 4, pp.5462-5480 (2022)

[13] **Rapid stellar and binary population synthesis with COMPAS**
Team COMPAS: Riley et al. (including Ryosuke Hirai)
The Astrophysical Journal Supplement Series, Volume 258, Issue 2, id.34, 30 pp. (2022)

[12] **COMPAS: A rapid binary population synthesis suite**
Team COMPAS: Riley et al. (including Ryosuke Hirai)
Journal of Open Source Software, vol. 7, issue 69, id. 3838 (2022)

[11] **Stellar Core-Merger-Induced Collapse: new Formation Pathways for Black Holes, Thorne-Żytkow objects, Magnetars and Superluminous Supernovae**
Iminhaji Ablimit, Philipp Podsiadlowski, Ryosuke Hirai, James Wicker
Monthly Notices of the Royal Astronomical Society, Volume 513, Issue 4, pp.4802-4813 (2022)

[10] **Supernova explosions in active galactic nuclear discs**
Evgeni Grishin, Alexey Bobrick, Ryosuke Hirai, Ilya Mandel, Hagai B. Perets
Monthly Notices of the Royal Astronomical Society, Volume 507, Issue 1, pp.156-174 (2021)

[9] **The observability of inflated companion stars after supernovae in massive binaries**
Misa Ogata, Ryosuke Hirai, Kotaro Hijikawa
Monthly Notices of the Royal Astronomical Society, Volume 505, Issue 2, pp.2485-2499 (2021)

[8] **Wind Mass-loss Rates of Stripped Stars Inferred from Cygnus X-1**
Coenraad J. Neijssel, Serena Vinciguerra, Alejandro Vigna-Gómez, Ryosuke Hirai, James C. A. Miller-Jones, Arash Bahramian, Thomas J. Maccarone, Ilya Mandel
The Astrophysical Journal, Volume 908, Issue 2, id.118, 9 pp. (2021)

[7] **Neutron Star Extreme Matter Observatory: A kilohertz-band gravitational-wave detector in the global network**
Ackley et al. (including Ryosuke Hirai)
Publications of the Astronomical Society of Australia, Volume 37, article id. e047 (2020)

[6] **The sensitivity of presupernova neutrinos to stellar evolution models**
Chinami Kato, Ryosuke Hirai, Hiroki Nagakura
Monthly Notices of the Royal Astronomical Society, Volume 496, Issue 3, pp.3961-3972 (2020)

[5] **A Subsolar Metallicity Progenitor for Cassiopeia A, the Remnant of a Type IIb Supernova**
Toshiki Sato, Takashi Yoshida, Hideyuki Umeda, Shigehiro Nagataki, Masaomi Ono, Keiichi Maeda, Ryosuke Hirai, John P. Hughes, Brian J. Williams, Yoshitomo Maeda
The Astrophysical Journal, Volume 893, Issue 1, id.49, 9 pp. (2020)

[4] **Origins of Type Ibn SNe 2006jc/2015G in interacting binaries and implications for pre-SN eruptions**
Ning-Chen Sun, Jusytn R. Maund, Ryosuke Hirai, Paul A. Crowther, Philipp Podsiadlowski
Monthly Notices of the Royal Astronomical Society, Volume 491, Issue 4, p.6000-6019 (2020)

[3] **Hydrodynamical simulations and similarity relations for eruptive mass loss from massive stars**
Stanley P. Owocki, Ryosuke Hirai, Philipp Podsiadlowski, Fabian R. N. Schneider
Monthly Notices of the Royal Astronomical Society, Volume 485, Issue 1, p.988-1000 (2019)

[2] **The W4 method: a new multi-dimensional root-finding scheme for nonlinear systems of equations**
Hirotada Okawa, Kotaro Fujisawa, Yu Yamamoto, Ryosuke Hirai, Nobutoshi Yasutake, Hiroki Nagakura, Shoichi Yamada
Applied Numerical Mathematics, Volume 183, p.157-172 (2023)

[1] **Formation pathway of Population III coalescing binary black holes through stable mass transfer**
Kohei Inayoshi, Ryosuke Hirai, Tomoya Kinugawa, Kenta Hotokezaka
Monthly Notices of the Royal Astronomical Society, Volume 468, Issue 4, p.5020-5032 (2017)

Other Articles

[8] **The Chaotic History of Eta Carinae**
Ryosuke Hirai
Proceedings of Stan Fest: Fundamentals of Stellar Outflows: Celebrating and Amplifying the Scientific Life of Stan Owocki, 241-249 (2026)

[7] **Common envelopes at StanFest**
Ilya Mandel, Ryosuke Hirai and Lewis Picker
Proceedings of Stan Fest: Fundamentals of Stellar Outflows: Celebrating and Amplifying the Scientific Life of Stan Owocki, 218-226 (2026)

[6] **Dusty Common Envelope Evolution**
Lionel Siess, Luis C. Bermúdez-Bustamante, Orsola De Marco, Daniel J. Price, Miguel Gonález-Bolívar, Chunliang Mu, Mike Y. M. Lau, Ryosuke Hirai and Taïssa Danilovich
XIV Torino Workshop on AGB stars Conference Proceedings, Galaxies, Volume 12, Issue 6, id.82 (2024)

[5] **Post-supernova binary interactions**
Ryosuke Hirai
41st Liège International Astrophysical Colloquium (LIAC41) Conference Proceedings (2024)

[4] **Roman CCS White Paper: Characterizing the Galactic population of isolated black holes**
C. Y. Lam et al. (including R. Hirai)
arXiv: 2306.12514

[3] **Science Cases for the Keck Wide-Field Imager**
J. Cooke et al. (including R. Hirai)
arXiv: 2207.11698

[2] **Common envelope simulations between a thermally pulsating AGB star and a low mass companion**
Miguel González-Bolívar, Orsola De Marco, Mike Y. M. Lau, Ryosuke Hirai, Daniel J. Price
American Astronomical Society Meeting Conference Proceedings #240, id. 425.02. Bulletin of the American Astronomical Society, Vol. 54, No. 6 e-id 2022n6i425p02

[1] **水素欠乏超新星の親星の起源(The Origin of the Progenitors of Stripped-Envelope Supernovae)**
平井 遼介 (Ryosuke Hirai)
天文月報*(The Astronomical Herald)*, Volume 111, Issue 9, p.580-588 (2018)

Invited Talks

2026.5 **Seminar**, *Kyoto University*, Japan.

2025.5 **Invited talk**, *Neutron star observations and theory - research stimulation workshop 2025*, Japan (online).

2025.5 **Seminar**, *University of Melbourne*, Australia.

2025.4 **Colloquium**, *JAXA*, Japan.

2025.2 **Invited talk**, *Theories of Astrophysical Big Bangs 2025*, Japan.

2024.11 **Colloquium**, *Academia Sinica Institute of Astronomy and Astrophysics*, Taiwan.

2024.11 **Invited talk**, *Stars in Brisbane*, Australia.

2024.8 **Invited talk**, *The Progenitors of Supernovae and their Explosions*, China.

2024.6 **Seminar**, *ABBL-iTHEMS, RIKEN*, Japan (online).

2024.5 **Seminar**, *RESCEU, University of Tokyo*, Japan.
2024.5 **Seminar**, *Department of Earth Science and Astronomy, University of Tokyo*, Japan.
2024.3 **Invited talk**, *Yamada lab 20th Anniversary Workshop*, Japan.
2023.10 **Seminar**, *Orange pulsar meeting (online)*, Australia.
2023.5 **Seminar**, *Orange pulsar meeting (online)*, Australia.
2022.11 **Invited talk**, *Supernovae in the Gravitational Wave Detection Era*, Australia.
2022.11 **Seminar**, *University of Melbourne*, Australia.
2022.10 **Seminar**, *University of Delaware (online)*, USA.
2022.9 **Seminar**, *Heidelberg Institute for Theoretical Studies*, Germany.
2022.9 **Seminar**, *Max Planck Institute for Astrophysics*, Germany.
2022.9 **Seminar**, *European Southern Observatory*, Germany.
2022.3 **Seminar**, *Rikkyo University*, Japan.
2022.3 **Seminar**, *Institute of Cosmic Ray Research, University of Tokyo*, Japan.
2021.6 **Seminar**, *Macquarie University (online)*, Australia.
2021.6 **Invited talk**, *European Astronomical Society Annual meeting 2021 (online)*, Netherlands.
2021.4 **Colloquium**, *Heidelberg (online)*, Germany.
2021.3 **Seminar**, *Technion (online)*, Israel.
2020.12 **Seminar**, *Hebrew University of Jerusalem (online)*, Israel.
2020.11 **Seminar**, *Kyoto University + Yukawa Institute (online)*, Japan.
2020.11 **Colloquium**, *SWIFAR, Yunnan University (online)*, China.
2020.11 **Colloquium**, *National Astronomical Observatory of Japan (online)*, Japan.
2020.10 **Invited talk**, *Stellar alchemy to galactic archaeology (online)*, Japan.
2020.5 **Seminar**, *RESCEU, University of Tokyo (online)*, Japan.
2020.4 **Colloquium**, *Monash University (online)*, Australia.
2020.2 **Invited talk**, *Phantom workshop*, Australia.
2019.11 **Colloquium**, *Tohoku University*, Japan.
2019.9 **Colloquium**, *JAXA*, Japan.
2019.9 **Seminar**, *Waseda University*, Japan.
2019.9 **Colloquium**, *University of Tokyo*, Japan.
2019.9 **Seminar**, *Peking University*, China.
2019.9 **Seminar**, *Yunnan National Astronomical Observatory*, China.
2019.4 **Seminar**, *Sheffield University*, UK.
2019.3 **Seminar**, *Anton Pannekoek Institute*, Netherlands.
2019.3 **Seminar**, *Heidelberg Institute of Theoretical Physics*, Germany.
2018.12 **Seminar**, *Chiba Institute of Technology*, Japan.
2018.10 **Seminar**, *Department of Earth Sciences, University of Oxford*, UK.
2018.9 **Seminar**, *University of Delaware*, USA.
2018.6 **Colloquium**, *Department of Physics, University of Oxford*, UK.

2018.1 **Seminar**, *Department of Physics, University of Oxford*, UK.
2017.12 **Seminar**, *Albert Einstein Institute*, Germany.
2017.11 **Seminar**, *Argelander Institute, University of Bonn*, Germany.
2017.10 **Colloquium**, *National Astronomical Observatory of Japan*, Japan.
2017.10 **Seminar**, *RIKEN*, Japan.
2017.5 **Colloquium**, *Institute of Cosmic Ray Research, University of Tokyo*, Japan.
2016.7 **Seminar**, *Kyoto University*, Japan.
2016.1 **Seminar**, *RESCEU, University of Tokyo*, Japan.
2015.7 **Seminar**, *KEK*, Japan.
2015.6 **Seminar**, *Kyoto University*, Japan.
2014.4 **Seminar**, *RIKEN*, Japan.

Contributed talks/posters

2026.3 **Contributed talk**, *Spring ASJ Annual meeting (online)*, Japan.
2025.8 **Contributed talk**, *Binary Stars in a New Era*, China.
2025.7 **Contributed talk**, *ASA Annual science meeting 2025*, Australia.
2025.6 **Contributed talk**, *Australian LSST Workshop 2025*, Australia.
2024.7 **Contributed talk**, *41st Liège International Astrophysical Colloquium*, Belgium.
2024.7 **Contributed talk**, *StanFest: Fundamentals of Stellar Outflows*, Belgium.
2024.3 **Contributed talk**, *Spring ASJ Annual meeting (online)*, Japan.
2024.2 **Contributed talk**, *ANITA workshop 2024*, Australia.
2023.12 **Contributed talk**, *RESCEU-NBIA workshop on gravitational-wave sources*, Japan.
2023.9 **Contributed talk**, *Australian LSST Workshop 2023*, Australia.
2023.3 **Contributed poster**, *eXtreme Black Holes*, USA.
2023.2 **Contributed talk**, *ANITA workshop 2023*, Australia.
2023.2 **Contributed talk**, *Phantom users workshop 2023*, Australia.
2022.12 **Contributed talk**, *Gravitational Wave Physics and Astronomy Workshop*, Australia.
2022.11 **Contributed talk**, *Supervirtual 2022*, Fully online.
2022.6 **Contributed talk**, *ASA Annual science meeting 2022*, Australia.
2022.3 **Contributed talk**, *Spring ASJ Annual meeting (online)*, Japan.
2021.12 **Contributed talk**, *34th Rironkon Symposium (online)*, Japan.
2021.10 **Contributed talk**, *Asymmetrical Post-main-sequence Nebulae 8 (online)*, Spain.
2021.9 **Contributed talk**, *Autumn ASJ Annual meeting (online)*, Japan.
2021.8 **Contributed talk**, *Neutron star workshop 2021 (online)*, Japan.
2021.7 **Contributed talk**, *ASA Annual science meeting 2021 (online)*, Australia.
2021.3 **Contributed talk**, *Spring ASJ Annual meeting (online)*, Japan.
2021.3 **Contributed poster**, *Triple Evolution and Dynamics (online)*, Israel.
2021.2 **Contributed talk**, *ANITA workshop 2021 (online)*, Australia.
2021.1 **Contributed talk**, *Binary/Variable star workshop 2020 (online)*, Japan.

2021.1 **Contributed poster**, *43rd COSPAR Assembly (online)*, Australia.
2020.2 **Contributed talk**, *ANITA workshop 2020 (online)*, Australia.
2019.9 **Contributed talk**, *Autumn ASJ Annual meeting*, Japan.
2019.5 **Contributed talk**, *FOE19 Fifty-one Erg*, USA.
2018.12 **Contributed poster**, *31st Rironkon Symposium*, Japan.
2018.11 **Contributed talk**, *Massive stars and supernovae*, Argentina.
2018.7 **Contributed talk**, *Frontiers of the Physics of Massive stars*, Mexico.
2017.9 **Contributed talk**, *Autumn ASJ Annual meeting*, Japan.
2017.7 **Contributed talk**, *Ringberg Workshop*, Germany.
2017.7 **Contributed poster**, *FOE17 Fifty-One Erg*, USA.
2017.3 **Contributed talk**, *Spring ASJ Annual meeting*, Japan.
2017.3 **Contributed talk**, *Ultraluminous X-ray source workshop*, Japan.
2017.1 **Contributed talk**, *Workshop on Transient Universe in the Big Survey Era*, Japan.
2016.12 **Contributed talk**, *29th Rironkon Symposium*, Japan.
2016.10 **Contributed talk**, *Binary/Variable star workshop 2016*, Japan.
2016.9 **Contributed talk**, *Autumn ASJ Annual meeting*, Japan.
2016.7 **Contributed talk**, *NAOJ-ECT workshop*, Japan.
2015.12 **Contributed poster**, *28th Rironkon Symposium*, Japan.
2015.9 **Contributed talk**, *Numazu workshop*, Japan.
2015.6 **Contributed talk**, *Fifty-One Erg*, USA.
2015.3 **Contributed talk**, *Spring ASJ Annual meeting*, Japan.
2014.12 **Contributed poster**, *27th Rironkon Symposium*, Japan.
2014.10 **Contributed talk**, *2nd DTA Symposium*, Japan.
2014.9 **Contributed poster**, *Binary systems, their evolution and environments*, Mongolia.
2014.9 **Contributed talk**, *Autumn ASJ Annual meeting*, Japan.
2014.2 **Contributed talk**, *Binary star workshop*, Japan.
2013.12 **Contributed poster**, *Multi-Messengers from Core-Collapse Supernovae*, Japan.
2013.9 **Contributed talk**, *Autumn ASJ Annual meeting*, Japan.

Service

Grant referee, *ISF (Israeli Science Foundation), ARC (Australian Research Council), DFG (German Research Foundation), National Science Centre Poland, Pazy Foundation*.

Journal referee, *MNRAS (Monthly Notices of the Royal Astronomical Society), PASJ (Publications of the Astronomical Society of Japan), Galaxies, ApJL (The Astrophysical Journal Letters), PASA (Publications of the Astronomical Society of Australia), ApJ (The Astrophysical Journal), A&A (Astronomy & Astrophysics), PRL (Physical Review Letters), Nature Communications, Ap&SS (Astrophysics and Space Science), PRD (Physical Review D), PASJL (Publications of the Astronomical Society of Japan Letters), PTEP (Progress of Theoretical and Experimental Physics)*.

2026 **External Panelist**, *XRISM proposal review*.

2025.3 **Conference organiser**, *Stellar black hole formation and detection*, SOC/LOC.

2024.6 **TA**, *MESA Summer School (MESA Down Under)*.

2024.2 **Conference organiser**, *Transients Down Under*, SOC.

2024– **Committee member**, *ANITA steering committee*.

2023– **Junior Associate representative**, *AAL Australian LSST Management Committee (ALManaC)*.

2021–2023 **External Panelist**, *Hubble Space Telescope proposal review*.

2021–2024 **Program chair**, *ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav), Relativistic Astrophysics program*.

2021.7 **Conference organiser**, *Nuclear burning in massive stars – towards the formation of binary black holes -*, (Monash hub leader).

2020–2022 **Seminar/Colloquia organiser**, *Monash University*.

2018–2019 **SPI-MAX seminar series organiser**, *University of Oxford*.

2015–2017 **Seminar organiser**, *Waseda University*.

Membership

2023– **Astronomical Society of Australia**.

2016– **Rironkon (Theoretical astrophysics society of Japan)**.

2015– **Astronomical Society of Japan**.

Outreach/Interdisciplinary events

2023.9 **Interdisciplinary Talk**, *Monash Science ECN Symposium 2023*.

2023.9 **Public event**, *Notting Hill Neighbourhood House Astronomy Night*.

2022.6 **Public Talk**, *Free Public Astronomy Lecture at Swinburne University of Technology*.

2022.5 **High School Talk**, *Melbourne High School*.

2021.8 **High School Talk**, *Hikawa High School*.

2021.3 **High School Talk**, *Waseda Jitsugyo High School*.

2019.7 **Public event**, *Cowley Road Carnival*.

2019.11 **Career advice session**, *U. Tokyo*.

2019.4 **Interdisciplinary Talk**, *Oxbridge Japanese Society meet-up*.

2019.1 **Public event**, *Stargazing Oxford 2019*.

2018.1 **Public event**, *Stargazing Oxford 2018*.

2016.3 **Interdisciplinary Talk**, *Early Bird Program Final Report Symposium*.

2015.6 **High School Talk**, *Waseda Jitsugyo High School*.